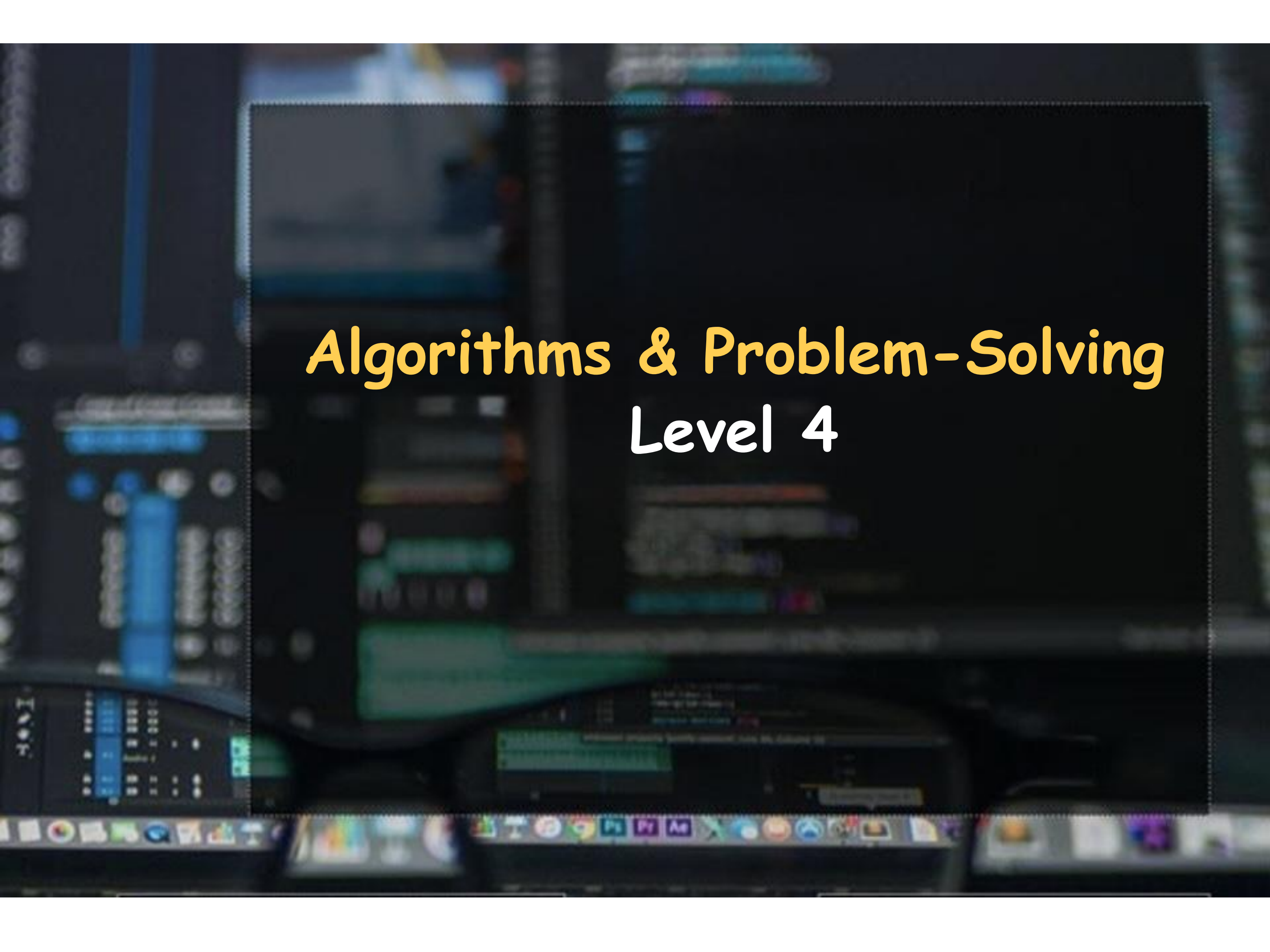
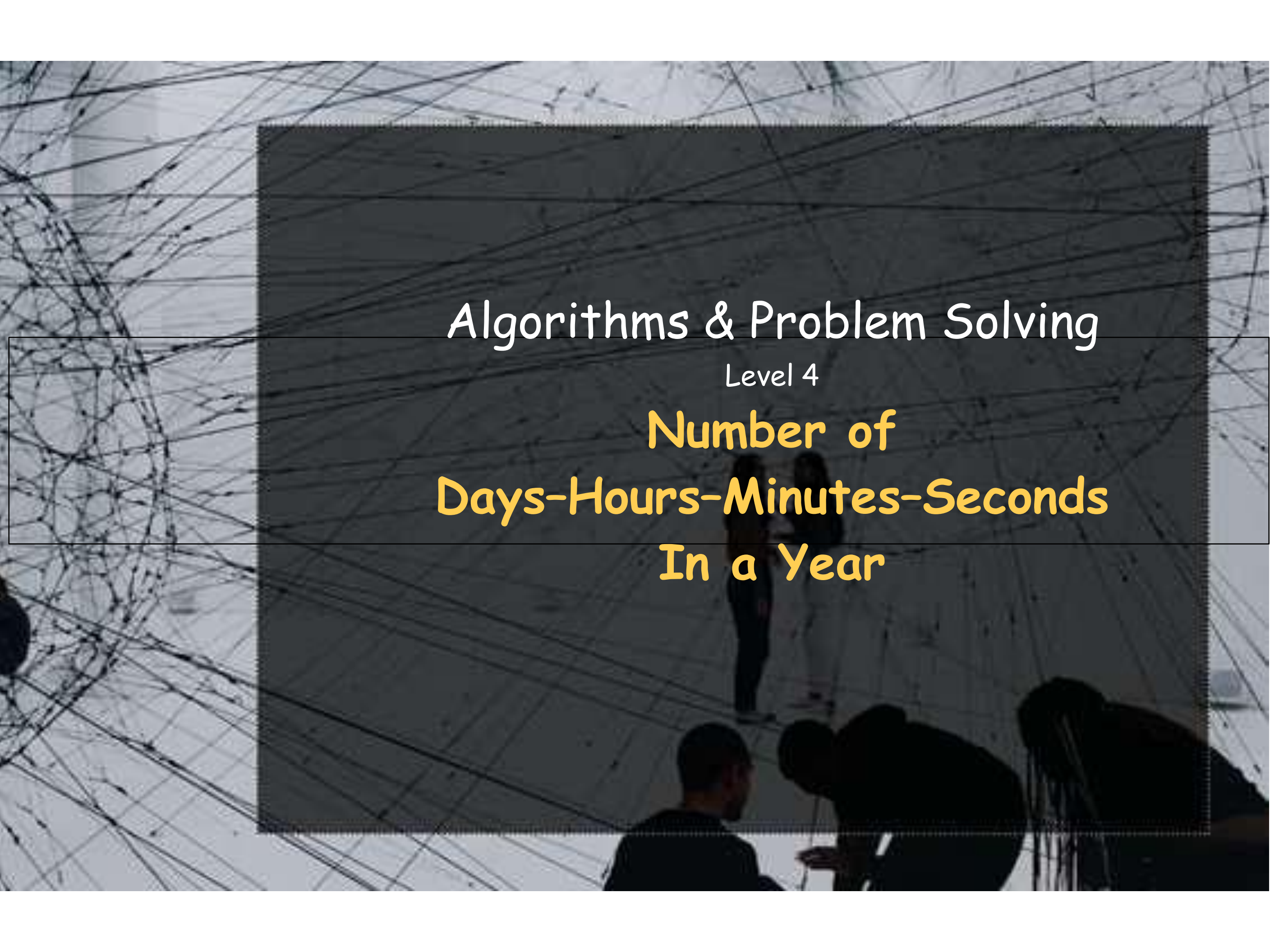




Algorithms & Problem-Solving Level 4

The background of the slide features a complex, abstract pattern of thin, dark lines crisscrossing over a lighter, textured surface. In the lower portion of the image, there are dark silhouettes of several people, some standing and some leaning over, suggesting a collaborative work environment or a public display. A large, dark, semi-transparent rectangular box is centered on the slide, serving as a backdrop for the text.

Algorithms & Problem Solving

Level 4

**Number of
Days-Hours-Minutes-Seconds
In a Year**



Definition

- The program's objective is to calculate the number of days, hours, minutes, and seconds in a given year, taking into account whether it is a leap year or not.



Explanation

- A leap year has 366 days, while a regular year has 365 days.
- The remaining values (hours, minutes, seconds) are calculated based on the number of days in a year.
 - The mathematical relationship between the units is:
 - 1 day = 24 hours
 - 1 hour = 60 minutes
 - 1 minute = 60 seconds

Procedures

① Read the year from the user:

- Using the `readYear()` function which prompts the user to enter the year and returns it as an integer

```
int readYear()
{
    int year;
    cout << "Please enter a year to check? ";
    cin >> year;
    return year;
}
```

② Checking if a year is a leap year:

- Using the `isLeapYear(int year)` function, which relies on the following mathematical rule:

```
return(year % 4 == 0 && year % 100 != 0) || (year % 400 == 0);
```

- That is, a year is a leap year if it:

- is divisible by 4
- is not divisible by 100
- or is divisible by 400

- 3 Calculating the number of days, hours, minutes, and seconds:
- The `printYearDetails(int year)` function is used as follows:
 `int days = isLeapYear(year) ? 366 : 365;`
 `int hours = days * 24;`
 `int minutes = hours * 60;`
 `int seconds = minutes * 60;`
 - The results are then displayed on the screen using `cout`.
- 4 Main Program:
- In `main()`, the following two functions are called:
 `int year = readYear();`
 `printYearDetails(year);`



Example:

Please enter a year to check? 2000

Number of Days in Year[2000] is 366

Number of Hours in Year [2000] is 8784

Number of Minutes in Year [2000] is 527040

Number of Seconds in Year [2000] is 31622400

Please enter a year to check? 1971

Number of Days in Year [1971] is 365

Number of Hours in Year [1971] is 8760

Number of Minutes in Year [1971] is 525600

Number of Seconds in Year [1971] is 31536000



Explanation

- If it's a leap year, an extra 24 hours (i.e., one day) are added to the total number of hours in the year, which is also reflected in the number of minutes and seconds.
- The calculations are sequential: each result depends on the previous one to ensure accuracy in the conversion between units.

Summary

- ✓ Leap year = 366 days → 8784 hours → 527040 minutes → 31622400 seconds
- ✓ Regular year = 365 days → 8760 hours → 525600 minutes → 31536000 seconds
- ✓ The program uses structured functions to calculate values accurately and clearly.

Thank you 😊